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United States Patent	12409740
Kind Code	B2
Date of Patent	September 09, 2025
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Electric work vehicle

Abstract

An electric work vehicle includes a first removable cover to cover a portion of a first harness exposed downward through a first opening, a second removable cover to cover a portion of a second harness exposed downward through a second opening, and a third removable cover to cover a portion of a motor exposed downward through a third opening.

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Appl. No.: 18/734667

Filed: June 05, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240317074 A1	Sep. 26, 2024

Foreign Application Priority Data

JP	2021-213186	Dec. 27, 2021
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Related U.S. Application Data

continuation parent-doc WO PCT/JP2022/038590 20221017 PENDING child-doc US 18734667

Publication Classification

Int. Cl.: B60L50/51 (20190101); **B60K1/00** (20060101)

U.S. Cl.:

CPC **B60L50/51** (20190201); B60K2001/006 (20130101); B60L2200/40 (20130101); B60L2210/12 (20130101); B60L2210/40 (20130101)

Field of Classification Search

CPC: B60L (50/51); B60L (2200/40); B60L (2210/12); B60L (2210/40); B60K (2001/006)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to Japanese Patent Application No. 2021-213186 filed on Dec. 27, 2021 and is a Continuation Application of PCT Application No. PCT/JP2022/038590 filed on Oct. 17, 2022. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to electric work vehicles.

2. Description of the Related Art

(2) A known example of an electric work vehicle is disclosed in JP 2021-953A. The electric work vehicle disclosed in JP 2021-953A includes a battery (“travel battery (4)”), an inverter (“inverter (14)”) configured to convert DC power from the battery to AC power, a motor (“motor (M)”)

driven by electric power supplied from the inverter, a travel device (“front wheels (10) and rear wheels (11)”) driven by the motor, and a DC-DC converter (“voltage converter (19)”) configured to step down electric power from the battery.

SUMMARY OF THE INVENTION

(3) In an electric work vehicle such as that as described above, usually, the battery and the DC-DC converter are connected to each other by a harness, and the inverter and the motor are connected to each other by a harness. For such an electric work vehicle, it is desired to facilitate the maintenance of the harnesses and the motor and to prevent the harnesses and the motor from being damaged by a flying stone from the ground.

(4) Under these circumstances, there is demand for an electric work vehicle that provides enhanced ease of maintenance of the harnesses and the motor and is configured to prevent the harnesses and the motor from being damaged by a flying stone from the ground.

(5) An example embodiment of the present invention provides an electric work vehicle including a battery, an inverter to convert DC power from the battery to AC power, a motor drivable by electric power supplied from the inverter, a travel device drivable by the motor, a DC-DC converter to step down electric power from the battery, a first harness connecting the battery and the DC-DC converter, a second harness connecting the inverter and the motor, a body frame including a first opening through which the first harness is exposed downward, a second opening through which the second harness is exposed downward, and a third opening through which the motor is exposed downward, and supporting the battery, the inverter, the motor, and the DC-DC converter, a first removable cover covering a portion of the first harness exposed downward through the first opening, a second removable cover covering a portion of the second harness exposed downward through the second opening, and a third removable cover covering a portion of the motor exposed downward through the third opening.

(6) With this characteristic configuration, even when a stone flies from the ground toward the first harness, the second harness, or the motor, the stone is blocked by the first cover, the second cover, or the third cover. Thus, the first harness, the second harness, and the motor can be prevented from being damaged by a flying stone from the ground.

(7) In addition, during maintenance, the first cover, the second cover, and the third cover are removed, and the first harness, the second harness, and the motor are thus exposed downward. As a result, it is easy to perform the maintenance of the first harness, the second harness, and the motor by accessing the first harness, the second harness, and the motor from below.

(8) Furthermore, in an example embodiment of the present invention, it is preferable that the motor is water-cooled, and the third cover covers a water-cooled portion of the motor.

(9) With this characteristic configuration, the water-cooled portion of the motor can be protected by the third cover.

(10) Furthermore, in an example embodiment of the present invention, it is preferable that at least any of the first cover, the second cover, or the third cover includes a drain hole.

(11) With this characteristic configuration, at least any of the first cover, the second cover, or the second cover allows water accumulating inside to be discharged through the drain hole.

(12) Furthermore, in an example embodiment of the present invention, it is preferable that a transmission shaft to transmit motive power from the motor to the travel device is provided, and the transmission shaft is held by at least any of the first cover, the second cover, or the third cover.

(13) With this characteristic configuration, at least any of the first cover, the second cover, or the second cover can be used as a support for the transmission shaft.

(14) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a left side view of a tractor.
- (2) FIG. 2 is a left side view showing arrangement of an inverter and the like.
- (3) FIG. 3 is a diagram showing a flow of motive power transmission.
- (4) FIG. 4 is a left side sectional view showing a configuration of a front portion of a body of the tractor.
- (5) FIG. 5 is a plan view showing the configuration of the front portion of the body of the tractor.
- (6) FIG. 6 is a bottom view showing the configuration of the front portion of the body of the tractor.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

(7) Example embodiments of the present invention will be described based on the drawings. In the following description, unless otherwise stated, the direction of an arrow F and the direction of an arrow B in the drawings will be respectively referred to as “the front side” and “the rear side”, and the direction of an arrow L and the direction of an arrow R in the drawings will be respectively referred to as “the left side” and “the right side”. Also, the direction of an arrow U and the direction of an arrow D in the drawings will be respectively referred to as “the upper side” and “the lower side”.

(8) The following describes a tractor according to the present example embodiment. As shown in FIG. 1, the tractor includes left and right front wheels **10**, left and right rear wheels **11**, and a cover **12**.

(9) The tractor also includes a body frame **2** and a driving section **3**. The body frame **2** is supported by the left and right front wheels **10** and the left and right rear wheels **11**.

(10) The cover **12** is disposed in a front portion of the body of the tractor. The driving section **3** is behind the cover **12**. In other words, the cover **12** is in front of the driving section **3**.

(11) The driving section **3** includes a protective frame **30**, a driver's seat **31**, and a steering wheel **32**. An operator can sit on the driver's seat **31**. Accordingly, the operator can get on the driving section **3**. The operator steers the left and right front wheels **10** by operating the steering wheel **32**. The operator can perform various driving operations in the driving section **3**.

(12) The tractor includes a battery **4**. The cover **12** is pivotable about an opening/closing axis Q extending in the left-right direction of the body. Accordingly, the cover **12** is openable and closable. When the cover **12** is closed, the battery **4** is covered by the cover **12**.

(13) As shown in FIG. 2, the tractor includes an inverter **14** and a motor M. The battery **4** supplies electric power to the inverter **14**. The inverter **14** converts DC power supplied from the battery **4** to AC power, and supplies the AC power to the motor M. The motor M is driven by the AC power supplied from the inverter **14**.

(14) As shown in FIGS. 2 and 3, the tractor includes a hydraulic continuously variable transmission **15** and a transmission **16**. As shown in FIG. 3, the hydraulic continuously variable transmission **15** includes a hydraulic pump **15a** and a hydraulic motor **15b**.

(15) The hydraulic pump **15a** is driven by rotational motive power transmitted from the motor M. As a result of the hydraulic pump **15a** being driven, rotational motive power is output from the hydraulic motor **15b**. The hydraulic continuously variable transmission **15** is configured to change the speed of rotational motive power between the hydraulic pump **15a** and the hydraulic motor **15b**. Also, the hydraulic continuously variable transmission **15** is capable of changing the transmission ratio in a stepless manner.

(16) The rotational motive power output from the hydraulic motor **15b** is transmitted to the transmission **16**. The speed of the rotational motive power transmitted to the transmission **16** is changed by a gear transmission mechanism included in the transmission **16**, and the rotational

motive power is distributed to the left and right front wheels **10** and the left and right rear wheels **11**. Thus, the left and right front wheels **10** and the left and right rear wheels **11** are driven.

(17) As shown in FIGS. **2** and **3**, the tractor also includes a middle PTO shaft **17** and a rear PTO shaft **18**. Rotational motive power output from the motor M is distributed to the hydraulic pump **15a**, the middle PTO shaft **17**, and the rear PTO shaft **18**. The middle PTO shaft **17** and the rear PTO shaft **18** are rotated by the distributed rotational motive power.

(18) If a work device is connected to the middle PTO shaft **17** or the rear PTO shaft **18**, the work device is driven by rotational motive power transmitted by the middle PTO shaft **17** or the rear PTO shaft **18**. For example, in the present example embodiment, a grass cutting device **19** is connected to the middle PTO shaft **17** as shown in FIG. **2**. The grass cutting device **19** is driven by rotational motive power transmitted by the middle PTO shaft **17**.

(19) As shown in FIGS. **4** to **6**, the motor M is behind the inverter **14**. The motor M and the inverter **14** are below the battery **4**. The motor M and the inverter **14** are aligned in the front-rear direction of the body. Specifically, the motor M and the inverter **14** are aligned in the front-rear direction of the body such that the inverter **14** is on the front side and the motor M is on the rear side. The tractor includes a radiator **20**, a DC (direct current)-DC converter **21**, and a battery **22**. The radiator **20**, the DC-DC converter **21**, and the battery **22** are in front of the battery **4**. The DC-DC converter **21** is to the immediate right of the radiator **20**. The battery **22** is to the immediate right of the DC-DC converter **21**.

(20) Here, all the inverter **14**, the motor M, and the DC-DC converter **21** are water-cooled, for example. The radiator **20** cools cooling water to cool the inverter **14**, the motor M, and the DC-DC converter **21**. A cooling hose **23** is connected to a lower portion of the radiator **20** and extends toward the inverter **14**. The DC-DC converter **21** steps down electric power supplied from the battery **4**. The battery **22** is to supply electric power to various electric components. The electric power (high voltage) supplied from the battery **4** is stepped down by the DC-DC converter **21** before being used to charge the battery **22** as low-voltage electric power.

(21) As shown in FIGS. **4** to **6**, a first harness **24** connecting the battery **4** and the DC-DC converter **21** extends from the battery **4** to the DC-DC converter **21**. The first harness **24** is to supply electric power from the battery **4** to the DC-DC converter **21**. An output section **4a** to which the first harness **24** is connected is provided at an upper portion of a front surface of the battery **4**. An input section **21a** to which the first harness **24** is connected is provided at a front portion of a lower surface of the DC-DC converter **21**. The first harness **24** is routed so as to extend downward from the output section **4a** and forward toward the input section **21a**.

(22) A second harness **25** connecting the inverter **14** and the motor M extends from the inverter **14** to the motor M. The second harness **25** is to supply electric power from the inverter **14** to the motor M. An output section **14a** to which the second harness **25** is connected is provided at a front portion of a lower surface of the inverter **14**. An input section Ma to which the second harness **25** is connected is provided at a front end of the motor M. The second harness **25** is routed so as to extend rearward from the output section **14a** toward the input section Ma.

(23) As shown in FIGS. **4** to **6**, the body frame **2** includes left and right side plates **40**, left and right bases **41**, a front plate **42**, a rear plate **43**, a bottom plate **44**, and a lateral plate **45**. The left and right side plates **40** are spaced apart from each other in the left-right direction of the body. A space between the left side plate **40** and the right side plate **40** is used as a space in which the motor M is disposed. The radiator **20**, the DC-DC converter **21**, and the battery **22** are supported on a base **46**. The base **46** is supported on front portions of the left and right side plates **40**.

(24) The left base **41** extends leftward from an upper edge of the left side plate **40**. The right base **41** extends rightward from an upper edge of the right side plate **40**. The inverter **14** is supported on the left and right bases **41**. The battery **4** is supported on a support stand **47**. The support stand **47** is supported on the left and right bases **41** and the rear plate **43**.

(25) The front plate **42** extends from a front end of the left side plate **40** to a front end of the right

side plate **40**. The rear plate **43** extends from a rear end of the left side plate **40** to a rear end of the right side plate **40**. The motor **M** is supported by the rear plate **43**. The bottom plate **44** extends from a lower edge of the left side plate **40** to a lower edge of the right side plate **40**. The lateral plate **45** is behind the bottom plate **44** and extends from the left side plate **40** to the right side plate **40**.

(26) A first opening **O1** is provided in a portion of the body frame **2** surrounded by the front plate **42**, a front edge of the bottom plate **44**, and the left and right side plates **40**. A second opening **O2** is provided in a portion of the bottom plate **44** below a rear portion of the second harness **25**. A third opening **O3** is provided in a portion of the body frame **2** surrounded by the rear plate **43**, a rear edge of the bottom plate **44**, and the left and right side plates **40**. That is, the body frame **2** has the first opening **O1**, the second opening **O2**, and the third opening **O3**.

(27) The first opening **O1** is configured to expose the first harness **24** and the cooling hose **23** in the downward direction. The second opening **O2** is configured to expose the second harness **25** in the downward direction. The third opening **O3** is configured to expose the motor **M** (specifically, a portion (water-cooled portion) **Mb** of the motor **M** in which cooling water flows) in the downward direction.

(28) As shown in FIGS. **4** to **6**, a first cover **48**, a second cover **49**, and a third cover **50** are attached to the body frame **2**. The first cover **48** is configured to cover the first opening **O1**. More specifically, the first cover **48** is configured to cover a portion of the first harness **24** exposed downward through the first opening **O1** and a portion of the cooling hose **23** exposed downward through the first opening **O1**. The portion of the first harness **24** exposed downward through the first opening **O1** and the portion of the cooling hose **23** exposed downward through the first opening **O1** are covered by the first cover **48** from below.

(29) The first cover **48** includes a horizontal surface portion **48a**, left and right bent portions **48b**, and recessed portions **48c**. The bent portions **48b** have shapes that are bent upward relative to the horizontal surface portion **48a** so as to extend along the respective side plates **40**. The recessed portions **48c** are configured to accommodate therein structures (e.g., a structure supporting a front portion of the grass cutting device **19**) around the first cover **48**, and thus, the first cover **48** does not obstruct the structures around the first cover **48**. The first cover **48** is fixed to the left and right side plates **40** by a plurality of bolts **51**. Specifically, each bent portion **48b** is pressed against the corresponding side plate **40** from the lateral outer side and is fixed to the side plate **40** by a plurality of (two in the present example embodiment) bolts **51**.

(30) The second cover **49** is configured to cover the second opening **O2**. More specifically, the second cover **49** is configured to cover a portion of the second harness **25** exposed downward through the second opening **O2**. The portion of the second harness **25** exposed downward through the second opening **O2** is covered by the second cover **49** from below.

(31) The second cover **49** includes a plurality of (for example, four in the present example embodiment) drain holes **49a**. The drain holes **49a** are provided in a portion of the second cover **49** that faces the second opening **O2**. The second cover **49** is pressed against the bottom plate **44** from below and is fixed to the bottom plate **44** by a plurality of (for example, four in the present example embodiment) bolts **52**.

(32) The third cover **50** is configured to cover the third opening **O3**. More specifically, the third cover **50** is configured to cover a portion of the motor **M** exposed downward through the third opening **O3**. In the present example embodiment, the third cover **50** is configured to cover the water-cooled portion **Mb** of the motor **M**. The portion (water-cooled portion **Mb**) of the motor **M** exposed downward through the third opening **O3** is covered by the third cover **50** from below.

(33) In the present example embodiment, the third cover **50** is configured to partially cover the third opening **O3**. A portion of the third opening **O3** between a front edge of the third cover **50** and the rear edge of the bottom plate **44** and a portion of the third opening **O3** between a rear edge of the third cover **50** and the rear plate **43** are not covered by the third cover **50**. The third cover **50** is

pressed against the lateral plate **45** from below and is fixed to the lateral plate **45** by a plurality of (for example, two in the present example embodiment) bolts **53**.

(34) Here, a transmission shaft **54** to transmit motive power from the motor M (motive power from the transmission **16**) to the left and right front wheels **10** is provided below the bottom plate **44**.

The transmission shaft **54** is held by the third cover **50** via a stay **55**.

Other Example Embodiments

(35) In the above-described example embodiments, the first cover **48** is configured to partially cover the first opening O1. However, the first cover **48** may be configured to cover the entire first opening O1.

(36) In the above-described example embodiments, the second cover **49** is configured to cover the entire second opening O2. However, the second cover **49** may be configured to partially cover the second opening O2.

(37) In the present example embodiment, the third cover **50** is configured to partially cover the third opening O3. However, the third cover **50** may be configured to cover the entire third opening O3.

(38) In the above-described example embodiments, the third cover **50** is configured to cover the water-cooled portion Mb of the motor M. However, the third cover **50** may be configured to cover a portion other than the water-cooled portion Mb of the motor M, in addition to the water-cooled portion Mb of the motor M.

(39) In the above-described example embodiments, the second cover **49** includes the four drain holes **49a**. However, the first cover **48** may include one to three, or five or more, drain holes **49a**.

(40) In the above-described example embodiments, the transmission shaft **54** is held by the third cover **50**. However, the transmission shaft **54** may be held by the second cover **49** instead of the third cover **50** or together with the third cover **50**.

(41) Example embodiments of the present invention are applicable to not only tractors but also combine harvesters, rice transplanters, and utility vehicles (multipurpose vehicles).

(42) While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. An electric work vehicle comprising: a battery; an inverter to convert DC power from the battery to AC power; a motor drivable by electric power supplied from the inverter; a travel device drivable by the motor; a DC-DC converter to step down electric power from the battery; a first harness connecting the battery and the DC-DC converter; a second harness connecting the inverter and the motor; a body frame including a first opening through which the first harness is exposed downward, a second opening through which the second harness is exposed downward, and a third opening through which the motor is exposed downward, and supporting the battery, the inverter, the motor, and the DC-DC converter; a first removable cover covering a portion of the first harness exposed downward through the first opening; a second removable cover covering a portion of the second harness exposed downward through the second opening; and a third removable cover covering a portion of the motor exposed downward through the third opening.
2. The electric work vehicle according to claim 1, wherein the motor is water-cooled; and the third cover covers a water-cooled portion of the motor.
3. The electric work vehicle according to claim 1, wherein at least any of the first cover, the second cover, or the third cover includes a drain hole.
4. The electric work vehicle according to claim 1, further comprising: a transmission shaft to

transmit motive power from the motor to the travel device; wherein the transmission shaft is held by at least any of the first cover, the second cover, or the third cover.
